第3章
一维问题

3.8 对流扩散方程
Advection-Diffusion Equation

寻找试探函数 $\theta(x)\in U$ 使得 $a(\theta,w)-F(w)=0 \quad \forall w\in U_0$

$$\theta^e=\boldsymbol{N}^e\boldsymbol{d}^e$$
$$w^e=\boldsymbol{w}^{e\mathrm{T}}\boldsymbol{N}^{e\mathrm{T}}$$
$$\boldsymbol{B}^e=\frac{\mathrm{d}\boldsymbol{N}^e}{\mathrm{d}x}$$

$$a(\theta,w)=\int_\Omega wAv\frac{\mathrm{d}\theta}{\mathrm{d}x}\mathrm{d}x+\int_\Omega\frac{\mathrm{d}w}{\mathrm{d}x}Ak\frac{\mathrm{d}\theta}{\mathrm{d}x}\mathrm{d}x=\sum_{e=1}^{n_{\mathrm{el}}}a^e(\theta^e,w^e)=\sum_{e=1}^{n_{\mathrm{el}}}\boldsymbol{w}^{e\mathrm{T}}\boldsymbol{K}^e\boldsymbol{d}^e$$

$$F(w)=\int_\Omega ws\,\mathrm{d}x-(Aw\bar{q})\big|_{\Gamma_q}=\sum_{e=1}^{n_{\mathrm{el}}}F^e(w^e)=\sum_{e=1}^{n_{\mathrm{el}}}\boldsymbol{w}^{e\mathrm{T}}\boldsymbol{f}^e$$

$$\boldsymbol{K}^e=a^e(\boldsymbol{N}^e,\boldsymbol{N}^{e\mathrm{T}})=\boldsymbol{K}_A^e+\boldsymbol{K}_D^e$$

$$\boldsymbol{f}^e=\int_{\Omega^e}\boldsymbol{N}^{e\mathrm{T}}s\mathrm{d}x-(A^e\boldsymbol{N}^{e\mathrm{T}}\bar{q})\big|_{\Gamma_q^e}$$

$$\boldsymbol{K}_A^e=\int_{\Omega^e}A^ev^e\boldsymbol{N}^{e\mathrm{T}}\boldsymbol{B}^e\mathrm{d}x \quad (\text{对流贡献})$$

$$\boldsymbol{K}_D^e=\int_{\Omega^e}A^ek^e\boldsymbol{B}^{e\mathrm{T}}\boldsymbol{B}^e\mathrm{d}x \quad (\text{扩散贡献})$$

对于两节点线性单元：

$$\boldsymbol{N}^e=\frac{1}{l^e}\begin{bmatrix}x_2^e-x & x-x_1^e\end{bmatrix} \qquad \boldsymbol{B}^e=\frac{1}{l^e}[-1\ \ 1]$$

$$\boldsymbol{K}_A^e=\frac{A^ev^e}{2}\begin{bmatrix}-1 & 1\\ -1 & 1\end{bmatrix} \qquad \boldsymbol{K}_D^e=\frac{kA^e}{l^e}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}$$

$$\boldsymbol{K}^e=\frac{kA^e}{l^e}\begin{bmatrix}1-P_e & -1+P_e\\ -1-P_e & 1+P_e\end{bmatrix} \qquad P_e=\frac{v^el^e}{2k}\ \text{(Peclet数)}$$

- 对流矩阵是非对称矩阵
- 系统矩阵不一定是正定的

  取 $\boldsymbol{z}^{\mathrm{T}}=[1,0]$

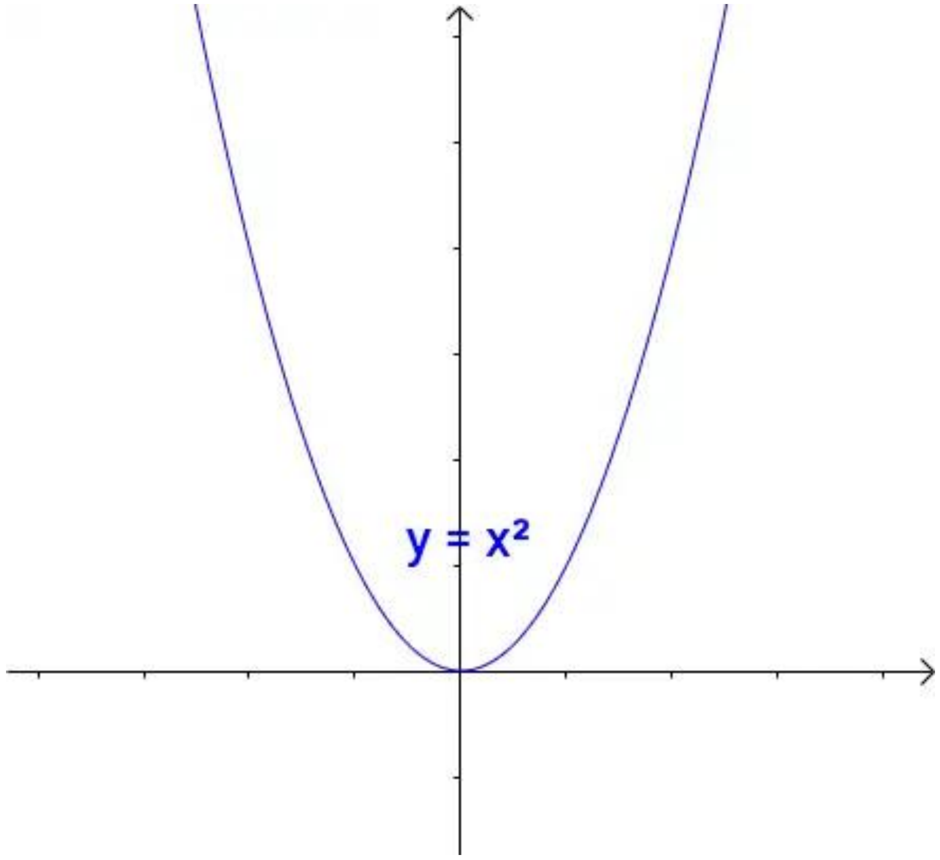
  $\boldsymbol{z}^{\mathrm{T}}\boldsymbol{K}^e\boldsymbol{z}=\dfrac{kA^e}{l^e}(1-P_e)<0$ (对流占优问题，$P_e>1$)
- 对称性和正定性的丧失使得方程求解及其困难

通过矩阵来研究二次函数(方程)，这就是线性代数 中二次型的重点。

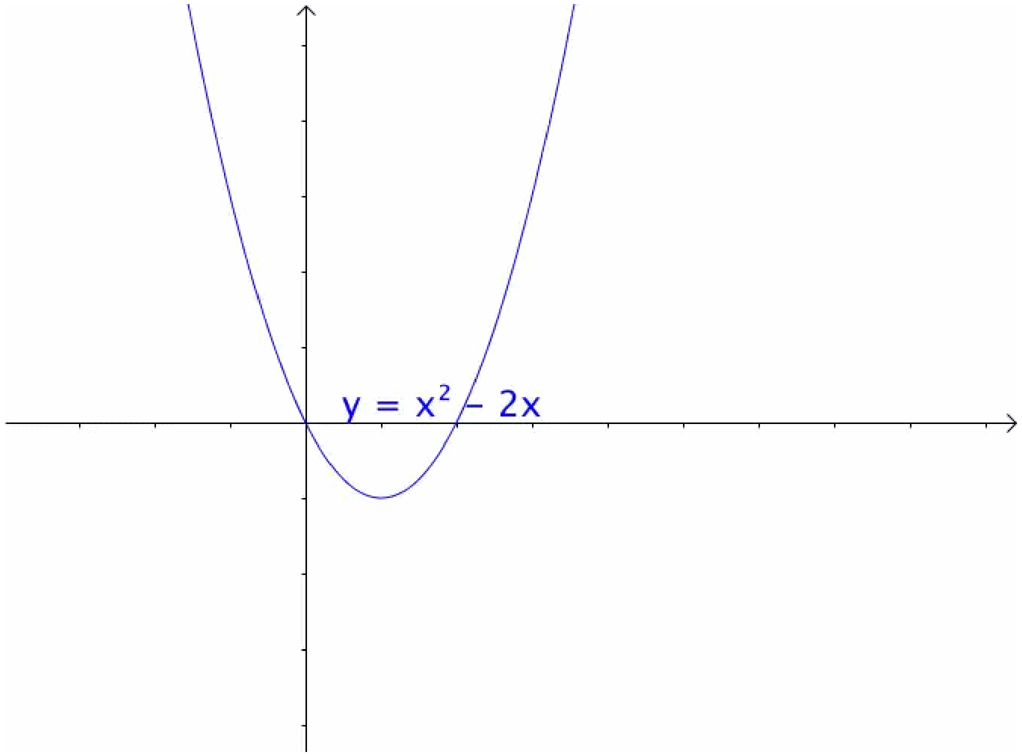
1 二次函数(方程)的特点

1.1 二次函数

最简单的一元二次函数就是:



给它增加一次项不会改变形状:



增加常数项就更不用说了，更不会改变形状。

3.8 对流扩散方程

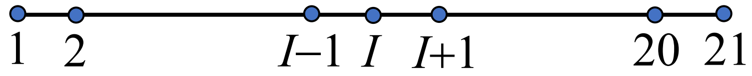
求解对流扩散方程

$$v\frac{\mathrm{d}\theta}{\mathrm{d}x}-k\frac{\mathrm{d}^2\theta}{\mathrm{d}x^2}=0$$

$$\theta(0)=0$$

$$\theta(10)=1$$

精确解：$\theta^{\mathrm{ex}}=\dfrac{\mathrm{e}^{vx/k}-1}{\mathrm{e}^{10v/k}-1}$



(20个等长线性单元)

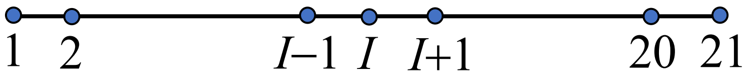
$$\boldsymbol{d}=[d_1\ \ d_2\ \ \cdots\ \ d_{I-1}\ \ d_I\ \ d_{I+1}\ \ \cdots\ \ d_{20}\ \ d_{21}]^{\mathrm{T}}$$

解答： 取$A^e=1$

$$\boldsymbol{K}^e\boldsymbol{d}^e=\boldsymbol{r} \qquad \boldsymbol{K}^e=\frac{k}{l^e}\begin{bmatrix}1-P_e & -1+P_e\\ -1-P_e & 1+P_e\end{bmatrix}$$

总体系统方程 $\boldsymbol{K}\boldsymbol{d}=\boldsymbol{r}$ $\qquad \boldsymbol{K}=?$

总体系统方程 $\boldsymbol{Kd} = \boldsymbol{r}$ $\quad \boldsymbol{K} = ?$



(20 linear elements with uniformly spaced nodes)

$$\boldsymbol{d} = [d_1 \quad d_2 \quad \cdots \quad d_{I-1} \quad d_I \quad d_{I+1} \quad \cdots \quad d_{20} \quad d_{21}]^{\mathrm{T}}$$

$$\boldsymbol{K}^e = \frac{k}{l^e}\begin{bmatrix} 1-P_e & -1+P_e \\ -1-P_e & 1+P_e \end{bmatrix} \xrightarrow[?]{} \boldsymbol{K}$$

$$\boldsymbol{K} = \frac{k}{l^e}\begin{bmatrix} 1-P_e & -1+P_e & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & -1-P_e & 2 & -1+P_e & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & -1-P_e & 1+P_e \end{bmatrix}$$

$$\begin{array}{cccccc} & e=1 & & e=I-1 & e=I & & e=20 \end{array}$$

$$\mathrm{LM} = \begin{bmatrix} 1 & \cdots & I-1 & I & \cdots & 20 \\ 2 & \cdots & I & I+1 & \cdots & 21 \end{bmatrix}$$

$$(1-P_e)d_1 + (-1+P_e)d_2 = r_1$$

$$(-1-P_e)d_{I-1} + 2d_I + (-1+P_e)d_{I+1} = 0 \quad 1 < I < 21$$

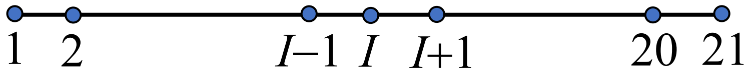
$$(-1-P_e)d_{20} + (1+P_e)d_{21} = r_{21}$$

求解对流扩散方程

$$v\frac{\mathrm{d}\theta}{\mathrm{d}x}-k\frac{\mathrm{d}^2\theta}{\mathrm{d}x^2}=0 \qquad \text{精确解：}\theta^{\mathrm{ex}}=\frac{\mathrm{e}^{vx/k}-1}{\mathrm{e}^{10v/k}-1}$$

$$\theta(0)=0$$

$$\theta(10)=1$$



(20个等长线性单元)

$$\boldsymbol{d}=[d_1\ \ d_2\ \ \cdots\ \ d_{I-1}\ \ d_I\ \ d_{I+1}\ \ \cdots\ \ d_{20}\ \ d_{21}]^{\mathrm{T}}$$

解答： 取$A^e=1$

$$\boldsymbol{K}^e\boldsymbol{d}^e=\boldsymbol{r} \qquad \boldsymbol{K}^e=\frac{k}{l^e}\begin{bmatrix}1-P_e & -1+P_e\\ -1-P_e & 1+P_e\end{bmatrix}$$

总体系统方程 $\boldsymbol{Kd}=\boldsymbol{r}$ 　 $\boldsymbol{K}=?$

$$(1-P_e)d_1+(-1+P_e)d_2=r_1$$

$$(-1-P_e)d_{I-1}+2d_I+(-1+P_e)d_{I+1}=0 \quad 2\le I\le 20$$

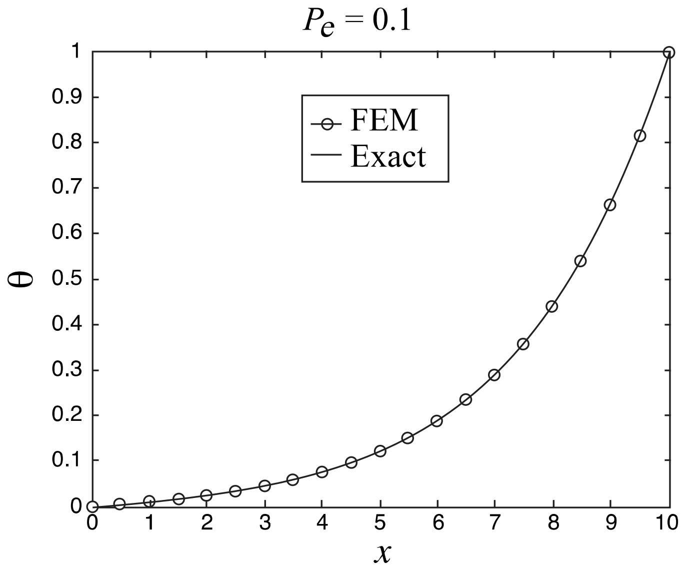
$$(-1-P_e)d_{20}+(1+P_e)d_{21}=r_{21}$$

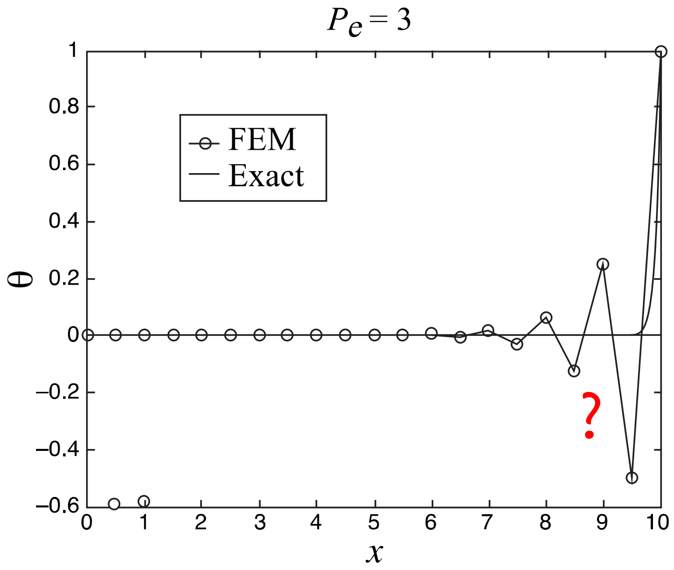
第I个方程的解：$d_I=\lambda^I$

$$(1-P_e)\lambda^2-2\lambda+(1+P_e)=0 \quad \text{(特征方程)}$$

$$\lambda_1=1,\ \lambda_2=\frac{1+P_e}{1-P_e} \qquad d_I=C_1+C_2\left(\frac{1+P_e}{1-P_e}\right)^I$$

$P_e>1$时解振荡！



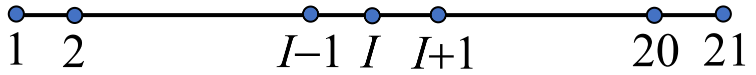


Advection_Diffusion.m

Advection_Diffusion-python (附录C.3)

- 当$P_e=0.1$时，有限元解精度很高
- 当$P_e=3$ 时，有限元解严重振荡 — 空间不稳定性
- 对于高Peclet数 (对流占优问题)，需要采用**特殊技术**以准确求解对流扩散方程

第I个内部节点的方程



$$(-1-P_e)d_{I-1}+2d_I+(-1+P_e)d_{I+1}=0 \qquad P_e=\frac{v^e l^e}{2k}$$

$$P_e(d_{I+1}-d_{I-1})-(d_{I+1}-2d_I+d_{I-1})=0$$

$$v^e\frac{d_{I+1}-d_{I-1}}{2l^e}-k\frac{d_{I+1}-2d_I+d_{I-1}}{(l^e)^2}=0$$

与中心差分格式相同

$$\left.\frac{\mathrm{d}\theta}{\mathrm{d}x}\right|_{x_I}\approx\frac{d_{I+1}-d_{I-1}}{2l^e} \qquad \left.\frac{\mathrm{d}^2\theta}{\mathrm{d}^2x}\right|_{x_I}\approx\frac{d_{I+1}-2d_I+d_{I-1}}{(l^e)^2}$$

$$\left[v\frac{\mathrm{d}\theta}{\mathrm{d}x}-k\frac{\mathrm{d}^2\theta}{\mathrm{d}x^2}\right]_I=0$$

解为什么振荡？

单向差分 (信息沿速度v的方向传播)：$\left.\frac{\mathrm{d}\theta}{\mathrm{d}x}\right|_{x_I}\approx\frac{d_I-d_{I-1}}{l^e}$　迎风差分近似

$$v^e\frac{d_I-d_{I-1}}{l^e}-k\frac{d_{I+1}-2d_I+d_{I-1}}{(l^e)^2}=0 \Leftarrow v^e\frac{d_I-d_{I-1}}{l^e}=\frac{v^e}{l^e}\left(\frac{d_{I+1}-d_{I-1}}{2}-\frac{d_{I+1}-2d_I+d_{I-1}}{2}\right)$$

人工扩散系数

$$v^e\frac{d_{I+1}-d_{I-1}}{2l^e}-\left(k+\frac{v^e l^e}{2}\right)\frac{d_{I+1}-2d_I+d_{I-1}}{(l^e)^2}=0$$

加辽金有限元解为什么振荡？
如何解决？

$$\frac{v^e l^e}{2k}(d_{I+1}-d_{I-1})-\left(1+\frac{v^e l^e}{2k}\right)(d_{I+1}-2d_I+d_{I-1})=0$$

$$-(1+2P_e)d_{I-1}+2(1+P_e)d_I-d_{I+1}=0$$

特征方程：$\lambda^2-2(1+P_e)\lambda+(1+2P_e)=0$

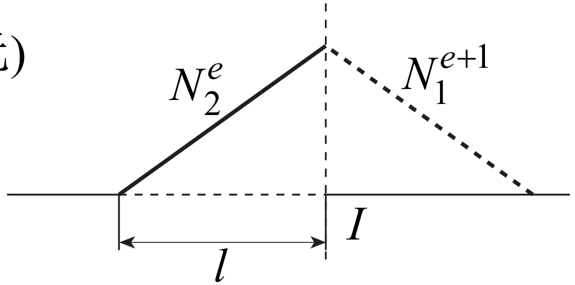
$$\lambda_1=1$$

$$\lambda_2=1+2P_e>0$$

$d_I=C_1+C_2(1+2P_e)^I$　无数值振荡！

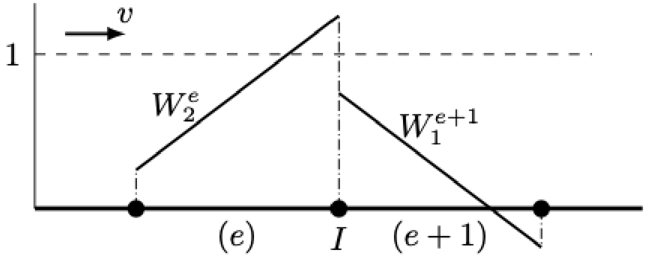
$$W_I^e = N_I^e + \alpha W_I^{e*} \qquad \int_{\Omega^e} W_I^{e*}\mathrm{d}x = \pm\frac{l^e}{2} \qquad W_I^{e*} = \frac{l^e}{2}\frac{\mathrm{d}N_I^e}{\mathrm{d}x}\mathrm{sgn}(v^e) \qquad \frac{\mathrm{d}N_I^e}{\mathrm{d}x} = \pm\frac{1}{l^e}\ \text{(线性单元)}$$

(选取方式不唯一)



$$\tilde{w}^e(x) = [\boldsymbol{N}^e + \frac{\alpha l^e}{2}\frac{\mathrm{d}\boldsymbol{N}^e}{\mathrm{d}x}\mathrm{sgn}(v^e)]\boldsymbol{w}^e$$

$$= w^e + \frac{\alpha l^e}{2}\frac{\mathrm{d}w^e}{\mathrm{d}x}\mathrm{sgn}(v^e) \qquad \frac{\mathrm{d}\tilde{w}^e}{\mathrm{d}x} = \frac{\mathrm{d}w^e}{\mathrm{d}x}\ \text{(线性单元)}$$



$$a^e(\theta^e, w^e) = \int_{\Omega^e}\tilde{w}^e A^e v^e \frac{\mathrm{d}\theta^e}{\mathrm{d}x}\mathrm{d}x + \int_{\Omega}\frac{\mathrm{d}\tilde{w}^e}{\mathrm{d}x}A^e k^e \frac{\mathrm{d}\theta^e}{\mathrm{d}x}\mathrm{d}x$$

$$\text{(线性单元)} = \int_{\Omega^e} w^e A^e v^e \frac{\mathrm{d}\theta^e}{\mathrm{d}x}\mathrm{d}x + \int_{\Omega} A^e k^e \left(1 + \frac{\alpha l^e |v^e|}{2k^e}\right)\frac{\mathrm{d}w^e}{\mathrm{d}x}\frac{\mathrm{d}\theta^e}{\mathrm{d}x}\mathrm{d}x$$

人工扩散

$$\boldsymbol{W}^e = \boldsymbol{N}^e + \frac{\alpha l^e}{2}\frac{\mathrm{d}\boldsymbol{N}^e}{\mathrm{d}x}\mathrm{sign}(v^e)$$

Streamline Upwind Petrov-Galerkin
流线迎风彼得罗夫-伽辽金法

$$\boldsymbol{K}^e = \boldsymbol{K}_A^e + (1 + \alpha P^e)\boldsymbol{K}_D^e$$

$$= \frac{k^e A^e}{l^e}\begin{bmatrix} 1 + (\alpha - 1)P_e & -1 - (\alpha - 1)P_e \\ -1 - (\alpha + 1)P_e & 1 + (\alpha + 1)P_e \end{bmatrix} \qquad P_e = \frac{v^e l^e}{2k^e}$$

$$\theta^e = \boldsymbol{N}^e\boldsymbol{d}^e \qquad w^e = \boldsymbol{w}^{e\mathrm{T}}\boldsymbol{N}^{e\mathrm{T}}$$

$$\boldsymbol{K}_A^e = \int_{\Omega^e} A^e v^e \boldsymbol{N}^{e\mathrm{T}}\boldsymbol{B}^e \mathrm{d}x$$

$$\boldsymbol{K}_D^e = \int_{\Omega^e} A^e k^e \boldsymbol{B}^{e\mathrm{T}}\boldsymbol{B}^e \mathrm{d}x$$

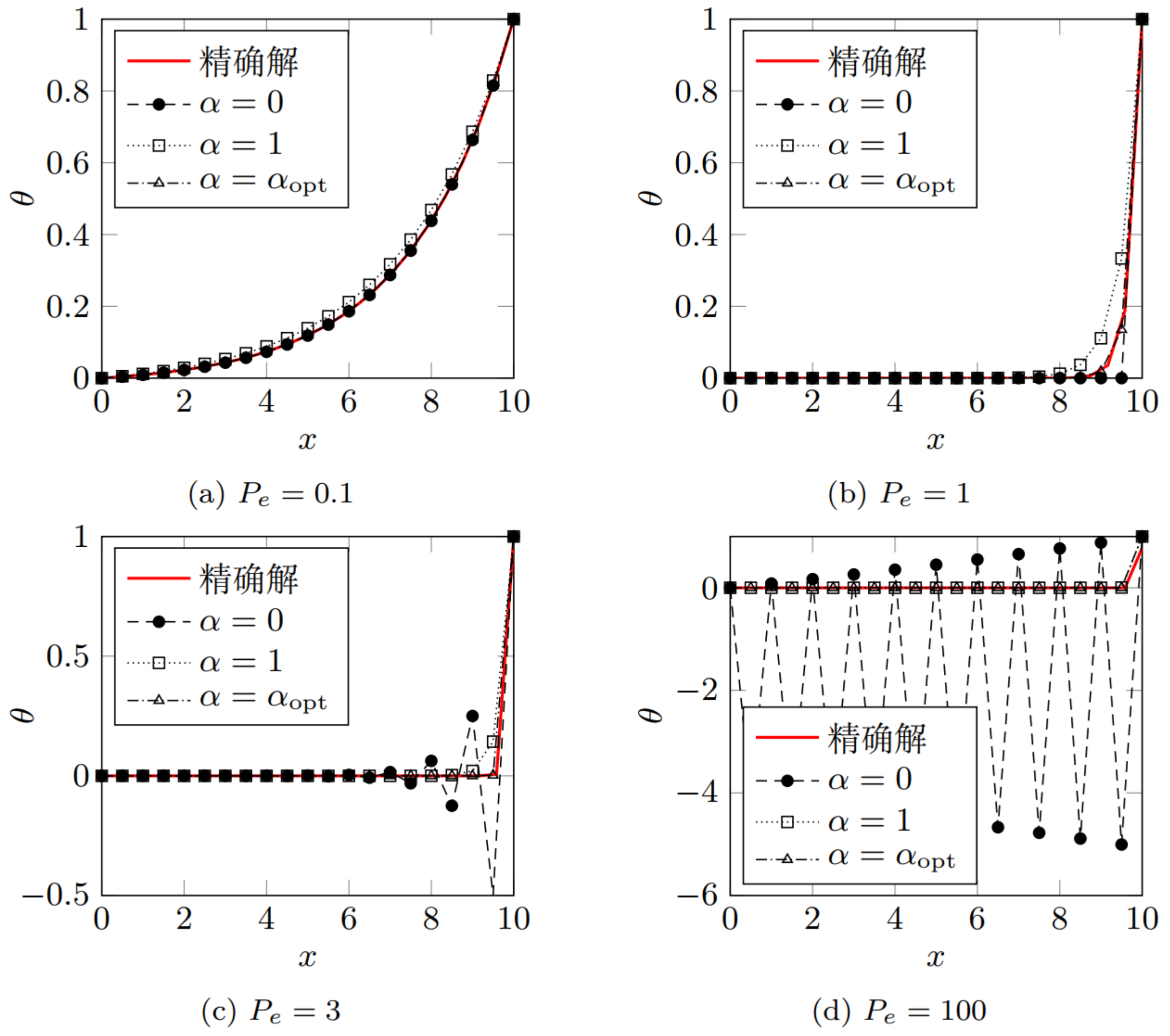
$$[-1 - (\alpha + 1)P_e]d_{I-1} + (2 + 2\alpha P_e)d_I + [-1 - (\alpha - 1)P_e]d_{I+1} = 0$$

$$\lambda_1 = 1,\ \lambda_2 = \frac{1 + (1 + \alpha)P_e}{1 - (1 - \alpha)P_e} \qquad d_I = C_1 + C_2\lambda^I$$

- $\alpha = 0$: Galerkin FEM
- $\alpha = 1$: One-side FD

- 当 $|\alpha| > \alpha_{\text{crit}} = 1 - 1/|P_e|$ 时，解不会振荡
- 当 $|\alpha| = \alpha_{\text{opt}} = \coth|P_e| - 1/|P_e|$ 时，对于任意 P_e 均给出节点精确解
- 伽辽金法当 $|P_e| > 1$ 时解振荡

彼得罗夫–伽辽金有限元法



- 伽辽金有限元在$P_e > 1$时解是振荡的
- 后向差分解虽不振荡，但由于引入了过量的人工扩散使得精度较差
- 而彼得罗夫伽辽金法当取$\alpha = \alpha_{opt}$时节点值是精确的

伽辽金最小二乘 (Galerkin Least Square) 法

$$\Pi^* = \Pi + \int_\Omega \tau \left[Av\frac{\mathrm{d}\theta}{\mathrm{d}x} - \frac{\mathrm{d}}{\mathrm{d}x}\left(Ak\frac{\mathrm{d}\theta}{\mathrm{d}x} \right) - s \right]^2 \mathrm{d}x$$

$$\delta\Pi^* = 0$$

小结

- 对流扩散方程

$$\int_\Omega wAv\frac{\mathrm{d}\theta}{\mathrm{d}x}\mathrm{d}x+\int_\Omega\frac{\mathrm{d}w}{\mathrm{d}x}Ak\frac{\mathrm{d}\theta}{\mathrm{d}x}\mathrm{d}x-\int_\Omega ws\,\mathrm{d}x+(Aw\bar{q})\Big|_{\Gamma_q}=0\quad\forall w$$

- 对于对流占优问题，伽辽金有限元解振荡
  - 从数学角度看：弱形式不适定（双线性形式不一定满足椭圆性条件） $a(\theta,w)=\int_\Omega(k\theta'w'+v\theta'w)\mathrm{d}x$
  - 从差分角度看：对流项导数采用中心差分近似，没有考虑上下游差异
  - 从弱形式看：伽辽金弱形式的权函数对称，没有考虑上下游差异
- 解决方案
  - 核心思想：引入人工扩散项，使得弱形式适定。引入的扩散项与单元长度l^e成正比，当l^e趋于零时人工扩散项趋于零
  - 差分法：迎风差分格式，尽可能多地利用上游传来的信息
  - 有限元法
    - ✓ 彼得洛夫-伽辽金弱形式，增加上游权重
    - ✓ 减小单元尺寸，使得 $P_e=\dfrac{v^e l^e}{2k}<1$